

# DRUM INSTRUMENT User Manual

Current: 2026-08-05

**Purpose:** Drum creation, sampling, sequencing, and impact placement. Create and shape drum material, then establish the musical impact and crowd-response field.

**Included products:** DIH Drum Morpher, BALLPARK, DrumKrushGlue

**Recommended route:** Drum route: DIH Drum Morpher -> Ballpark -> DrumKrushGlue.

**Preset law:** Factory presets describe musical intent and expose the useful range of the plugin. They do not change protected Input, Mix, Output, Power, or final ceiling controls unless the plugin's explicit canon requires it. Cathedral and SacredVerb retain their protected preset behavior. Preset changes must remain clickless.

## DIH Drum Morpher

Sixteen-pad drum sampler and host-synced sequencer with sample loading, pad shaping, restored factory programs, pattern memory, capture, morph, and the original DIH Layer Engine. Each pad can keep its sample unchanged, combine it with a modeled drum layer, or move toward a synthesized drum voice.

**Preferred placement:** Instrument Track / Drum Source. Load as a drum instrument, then route its output through Ballpark and DrumKrushGlue when that chain serves the production.

### Core controls

- 16 velocity-sensitive pads and MIDI notes
- WAV, AIFF, and CAF sample loading
- Volume, pitch, morph, filter, punch, air, and ADSR per pad
- Per-pad Sample, Hybrid, and Synth engine modes
- Per-pad Layer amount and controlled repeat-hit Variation
- Host tempo, swing, grid rate, steps, play, and record
- Global morph, metamorphosis, momentum bend, governor, and ceiling
- Sampler capture, playback, and transfer to pad
- Sixteen per-pad mute toggles for isolating or removing kit voices

### Sidechain and wave guide

- Drum Morpher is an instrument and does not require a host sidechain. CAPTURE can record an incoming source and transfer it to any pad.
- The current Audio Unit includes the restored factory program bank and uses a prepared, per-sample output-normalization ramp for smooth level transitions.

### Layer Engine quick guide

| Mode   | What it does                                                            | Best starting use                                                                 |
|--------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| SAMPLE | Plays only the loaded sample through Drum Morpher shaping               | When the source already has the exact identity you want.                          |
| HYBRID | Keeps the sample and blends an original DIH physical/model layer        | Default choice for more body, definition, or repeat-hit life.                     |
| SYNTH  | Keeps a quieter sample imprint while bringing the modeled layer forward | Electronic kicks, tuned toms/808 support, synthetic percussion, and sound design. |

- **LAYER** controls how much modeled reinforcement is added. Start around 15-30 percent. Increase by role: kick and toms can take more; hats and cymbals usually need less.

- **VARIATION** introduces small, deterministic changes in pitch, tone, and stereo position for repeated hits. Start around 5-15 percent. It does not randomize the pattern or replace velocity.
- MIDI velocity still controls hit level. The modeled layer also becomes brighter and more forceful at higher velocities, so programming velocity gives the kit a played response.
- Kick, snare, clap, rim, hats, hand percussion, tom/808, crash, and ride pads use different original DIH models. No Reason Studios samples or presets are included.

### Meter and visualizer guide

| Display           | What it measures                                     | How to read it                                                   |
|-------------------|------------------------------------------------------|------------------------------------------------------------------|
| L / R output      | Left and right output activity                       | Confirms stereo drum output and imbalance.                       |
| GOVERNOR          | Output containment activity                          | More activity means protection is holding energy back.           |
| MORPH             | Current metamorphosis activity                       | Shows how strongly the active pads are transforming.             |
| Pad waveform      | Loaded sample amplitude over time                    | Use to identify attacks, tails, and silence in each pad.         |
| Layer readout     | Amount of modeled reinforcement for the selected pad | Raise until the drum gains substance, then bypass-match.         |
| Variation readout | Repeat-hit pitch, tone, and stereo movement          | Use low values for realism; higher values for electronic motion. |

Start from reset or a purpose-matched preset, bypass-match, and adjust one control at a time.

## BALLPARK

Drum impact and placement processor whose base progression and hit calls turn performance energy into an audible stadium-response field.

**Preferred placement:** Drum Track / Drum Bus. Use as a drum instrument and impact-placement stage before DrumKrushGlue.

### Core controls

- Base mode and crowd-response progression
- Mound gain and switch-hitter transient mode
- Drive, depth, width, focus, clarity, mix, and convergence
- Input, RMS, peak, gain-reduction, and ceiling meters
- Foul, walk, single, double, triple, and home-run hit calls

### Meter and visualizer guide

| Display                  | What it measures                             | How to read it                                                             |
|--------------------------|----------------------------------------------|----------------------------------------------------------------------------|
| IN                       | Input level                                  | Confirms drum energy entering the stadium model.                           |
| GR                       | Gain reduction                               | Shows active impact containment.                                           |
| OUT                      | Output level                                 | Use with IN for level matching.                                            |
| RMS / PEAK               | Body level and fast peak                     | Their separation describes drum crest and punch.                           |
| Ceiling score / hit call | Composite impact, control, and ceiling score | Walk through Home Run describes response intensity, not sports prediction. |

### Current factory programs detected

- ON; GAN

Start from reset or a purpose-matched preset, bypass-match, and adjust one control at a time.

# DrumKrushGlue

Bus-preferred drum cohesion stage for binding hits into one controlled impact field, with an optional kick-key sidechain that opens a pocket only in its added weight and glue layer.

**Preferred placement:** Drum Bus / Parallel Drum Impact Return. Use on the drum bus or a parallel drum-impact return after Drum Morpher and Ballpark when the kit needs adhesive impact.

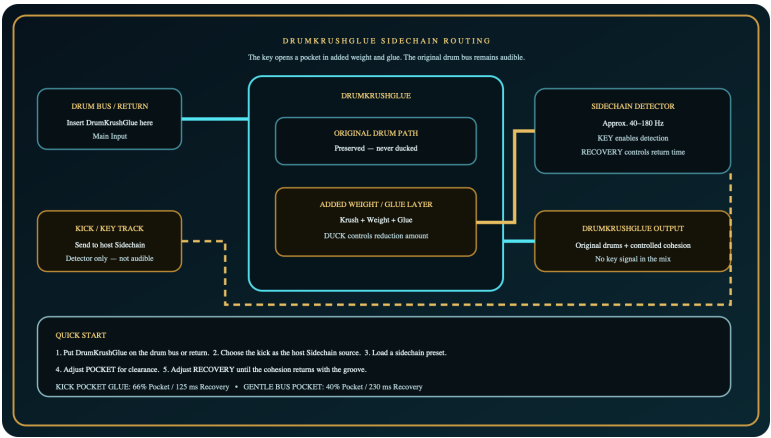
## Core controls

- Power
- Gain Match
- Sidechain Key
- Pocket Duck
- Key Recovery

## Sidechain and wave guide

- Route a kick or other guide to the host's named Sidechain input, then enable KEY. The sidechain controls the pocket detector only and is never mixed into the drum bus.
- POCKET moves only the added Krush/Weight/Glue contribution aside; RECOVERY controls how quickly cohesion returns. The original drum path remains audible.
- KICK POCKET GLUE loads 66 percent Pocket / 125 ms Recovery. GENTLE BUS POCKET loads 40 percent Pocket / 230 ms Recovery. All earlier programs keep KEY off.

## Sidechain routing schematic



DrumKrushGlue sidechain routing

The dashed gold route is detector control only; it never enters the audible output.

## Meter and visualizer guide

| Display | What it measures                         | How to read it                                                 |
|---------|------------------------------------------|----------------------------------------------------------------|
| IMPACT  | Detected drum peak activity              | Shows hit strength entering the impact path.                   |
| BODY    | Sustained drum energy                    | Shows weight carried between hits.                             |
| REPAIR  | Glue/control activity                    | Shows how strongly cohesion is correcting the bus.             |
| CEILING | Protected output pressure                | Persistent maximum activity means the chain is overdriven.     |
| KEY     | Detected 40-180 Hz sidechain energy      | Moves only when KEY is enabled and a host sidechain is routed. |
| POCKET  | Reduction of the added weight/glue layer | The original drum path remains outside this reduction.         |

Start from reset or a purpose-matched preset, bypass-match, and adjust one control at a time.